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The Impact of a Methicillin-Resistant Staphylococcus Aureus Nasal Polymerase Chain Reaction Protocol on Vancomycin Length of Therapy Among Patients With Skin and Soft Tissue Infections

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Abstract

Objective: We evaluated the impact of a methicillin-resistant Staphylococcus aureus (MRSA) nasal polymerase chain reaction (PCR) protocol on the vancomycin length of therapy (LOT) for skin and soft tissue infections (SSTIs). **Design:** Retrospective quasi-experimental pre- and post- MRSA nasal PCR protocol implementation study. **Setting:** Tertiary-care academic medical center in Jacksonville, Florida. **Patients:** Eligible patients received empiric vancomycin for SSTIs from January 1st to September 30th 2020 (pre-implementation group) and from January 1st to September 30th 2022 (post-implementation group). **Intervention:** The electronic health system software was modified to provide a best-practice advisory (BPA) prompt to the pharmacist upon order verification of vancomycin for patients with SSTIs. **Methods:** We reviewed patient records to determine the time from vancomycin prescription to de-escalation. The secondary outcomes were incidence of acute kidney injury (AKI), number of vancomycin levels collected, and hospital length of stay (LOS). **Results:** The study included 131 patients (pre-implementation, n = 86 and post-implementation, n = 45). There was no significant difference in vancomycin length of therapy (LOT) between implementation groups: mean LOT in days and standard deviation (SD) were 2.7 (1.9) and 2.6 (1.3), respectively, p-value 0.493. Of significance, in the post-implementation group, vancomycin LOT between patients with a negative and positive MRSA PCR were 2.3 (1.1) and 3.9 (1.6), p-value 0.006. There was no difference in secondary outcomes. **Conclusion:** The utilization of the MRSA nasal PCR to guide vancomycin de-escalation did not significantly change the vancomycin LOT, however in the post-implementation group there was a significant difference in vancomycin LOT between negative and positive MRSA PCRs.

Keywords

antimicrobial stewardship, skin and soft tissue infection, MRSA nares

Introduction

Skin and soft tissue infections (SSTIs) are a leading etiology of bacterial infections in the United States.[1] Methicillin-resistant *Staphylococcus aureus* (MRSA) is one of the most frequently isolated antibiotic resistant pathogens that cause SSTIs and is associated with high mortality rates and increased health care costs.[2] The 2014 Infectious Disease Society of America (IDSA) Practice guidelines for the diagnosis and management of SSTIs recommends anti-MRSA therapy in patients with carbuncles or abscesses with the following criteria: patients who have failed initial antibiotic treatment, patients who have markedly impaired host defenses, or patients meeting systemic inflammatory response syndrome criteria.[1] However, this guideline offers minimal guidance on approaches to anti-MRSA de-escalation. In many cases, high-quality skin and soft-tissue cultures are either unavailable or the finalized microbiological cultures have not resulted in growth of a

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causative pathogen, and clinicians are faced with the decision of when anti-MRSA coverage can be safely de-escalated. There is a desperate need for strategies to improve antibiotic utilization in contemporary health care settings, as antibiotic resistance has become a significant threat of the current era. One approach in the diagnosis and management of infectious diseases is the use of rapid molecular tests. These tests can markedly influence patient care by enabling timely de-escalation of antimicrobial therapy.

MRSA commonly colonizes the anterior nares, and assay-confirmed colonization has been recognized as a conceivable predictor of future clinical infection.[3] Multiple recent studies have evaluated the clinical utility of MRSA nasal screening in predicting clinical MRSA infection in SSTI. Emerging data has demonstrated robust negative predictive value (NPV) results, anywhere from 89.6% - 97.5%.[4-7] It is important to continue validating the high NPV in SSTIs to support possible de-escalation of antimicrobials, particularly vancomycin. In SSTIs, if the causative pathogen is not identified, de-escalation of vancomycin can be difficult, which can contribute to longer duration of vancomycin therapy, which can increase the risk of *Clostridioides difficile* infection, increase the risk of adverse events including acute kidney injury (AKI), Stevens-Johnson Syndrome, thrombocytopenia, vancomycin flushing reaction, rate of resistance, as well as resource utilization costs such as nursing administration and therapeutic drug monitoring.[8] The aim of this analysis is to determine the clinical utility of using the MRSA nasal polymerase chain reaction (PCR) screen to de-escalate vancomycin in patients with SSTIs.

Materials and Methods

MRSA PCR Protocol Development and Implementation

Based on the strong body of literature supporting the NPV of MRSA nasal screening in establishing the absence of MRSA in SSTIs, a MRSA nasal PCR protocol was developed as a part of an antimicrobial stewardship program at Mayo Clinic. This protocol included a Best Practice Advisory (BPA) prompt in the electronic health system software that prompted pharmacists to order a MRSA nasal PCR when a new order was placed for intravenous vancomycin for the indication of SSTI. MRSA nasal PCR test results were performed in-house 24 hours a day, seven days a week and resulted within two hours once obtained by the micro-lab using the GeneXpert MRSA system (Cepheid, Sunnyvale, CA). If the MRSA nasal PCR was negative, the pharmacist would reach out to the primary team to initiate discussion about vancomycin de-escalation. Education regarding the protocol was provided to the medical staff and pharmacy on August 29th, 2022. The BPA prompt was implemented in the electronic health system software on September 1st, 2022.

Study Design and Patients

This was a retrospective quasi-experimental pre- and post-MRSA nasal PCR protocol implementation study that was conducted at Mayo Clinic in Florida, a 304-bed, nonprofit, academic medical center. This study was approved by the Institutional Review Board. Cases were identified by generating a report of patients on vancomycin with an indication of SSTI using Slicer Dicer in EPIC. Patients were included if they received more than one dose of intravenous vancomycin empirically for SSTI indication within 48 hours of admission. Patient data in the pre-implementation group was collected from January 1st to September 30th 2020 and from January 1st to September 30th 2022 in the post-implementation group. Patients were excluded if they had a prosthesis near the site of SSTI, a concomitant active diagnosis of osteomyelitis, septic arthritis, necrotizing fasciitis, received nasal mupirocin at any point during the hospital stay, had an order for a MRSA nasal PCR that was never collected or resulted, or required vancomycin for a concomitant infection other than SSTI. Patients were excluded if they had a MRSA nasal screen obtained pre-implementation as we did not want the result to influence vancomycin LOT.

Data Collection

Data was collected from the electronic medical record (EMR) and the admission demographic information included age, sex, weight, height, antibiotic allergies, prior history of MRSA within the last 12 months and hospitalization within the last 30 days. Additionally, prior history of intravenous and oral antibiotics in the last 90 days was collected. Other data obtained included serum creatinine, maximum temperature on day of admission and 24 hours after the last dose of vancomycin administered. Cultures were collected, including source of specimen, whether skin cultures were obtained in the operating room, whether there was bacterial growth from skin cultures, and type of skin and soft tissue cultures. Cultures collected in the operating room were categorized as surgical cultures and all other cultures were classified as non-surgical. Dates of admission and discharge and whether the infectious diseases team was consulted was also collected.

Study Outcomes

The primary outcome of this study was intravenous vancomycin LOT in days from the start of the first dose to administration time of the last dose. The secondary outcomes were incidence of AKI, number of vancomycin levels collected, and hospital LOS.

AKI was defined as the doubling of serum creatinine or increase in serum creatinine by 0.5 mg/dL or higher after the first dose of vancomycin, per RIFLE criteria.[9] Serum creatinine levels were followed daily during vancomycin administration day(s) and up to 48-hour after discontinuation of vancomycin. Initial serum creatinine was obtained prior to the first intravenous vancomycin dose. If the serum creatinine was not available prior to the first dose, the first serum creatinine available after the first dose was recorded. Vancomycin levels were collected between the first and last vancomycin doses. Hospital LOS was determined from the admission date to the discharge date.

Statistical Analysis

Study power was determined to be 77% based on a standard deviation (SD) of 2 and an effect size of half a day to obtain a reduction in vancomycin LOT by one day between the pre- and post-implementation groups. Based on a two-sample *t* test with pool variances, 95 patients pre-intervention and 48 patients post-intervention were needed, assuming a 2:1 ratio between pre- and post-intervention patients. Continuous and nonparametric data were analyzed using the Kruskal-Wallis test and categorical data was analyzed using the chi-squared test. All tests were two-sided with an a priori alpha level of 0.05 for significance. Study data were collected and managed using REDCap hosted at the Mayo Clinic in Jacksonville, Florida and analyzed using BlueSky Statistics software (BlueSky Statistics LLC, Chicago, IL, USA).[10,11]

Results

Demographics

In total, 231 records were reviewed, and 100 patients were excluded, leaving 131 patients included in the study. There were 86 patients in the pre-implementation group and 45 patients in the post-implementation group (Figure 1). Baseline demographic data were similar except hospitalization within the last 30 days of admission and receipt of antibiotics in the last 90 days. Hospitalization within the last 30 days of admission was higher in the pre-implementation group vs the post-implentation group [15 (17.6%) vs 1 (2.2%)], p-value 0.010. Prior antibiotic use within the last 90 days was higher in the post-implementation group as compared with the pre-implementation group [26 (57.8%) vs 25 (29.1%)], p-value < 0.001 (Table 1). There was a significant difference in anti-MRSA agent usage within 90 days of admission between the pre- and post-implementation groups [12 (14.0%) and 14 (31.1%), p-value 0.019].

In terms of clinical characteristics, there was no difference between the pre- and post-implementation groups regarding max temperature on admission, temperature 24 hours after the last dose of vancomycin or classification of SSTI (Table 2). The Infectious Diseases team was consulted more frequently in the pre-as compared with post-implementation group [34 (39.5%) vs 8 (17.8%)], p-value 0.011. Cultures were collected in 83 (96.5%) patients in the pre-implementation and 42 (93.3%) in the post-implementation group. In the pre-implementation group, 50 skin cultures were obtained, of which 9 grew MRSA, 6 cultures *Corynebacterium* and 1 culture Methicillin-Resistant *Staphylococcus epidermis* (MRSE). In the post implementation group, 17 skin cultures were obtained, of which 7 grew MRSA and 4 grew *Corynebacterium*. There was no difference in bacterial growth in cultures between groups. In the pre-implementation group, one patient had MRSA bacteremia with no MRSA bacteremias in the post-implementation group.

Our study found that the most common site of infection was the lower extremities, 29 (33.7%) in the pre-group and 25 (55.6%) in the post-group, followed by upper extremity location of infection. In the pre-implementation group, there were significantly higher trunk infections compared to the post-group [21 (24.4%) vs 4 (8.9%)]. The most common type of infection was non-purulent cellulitis [44 (51.2%) vs

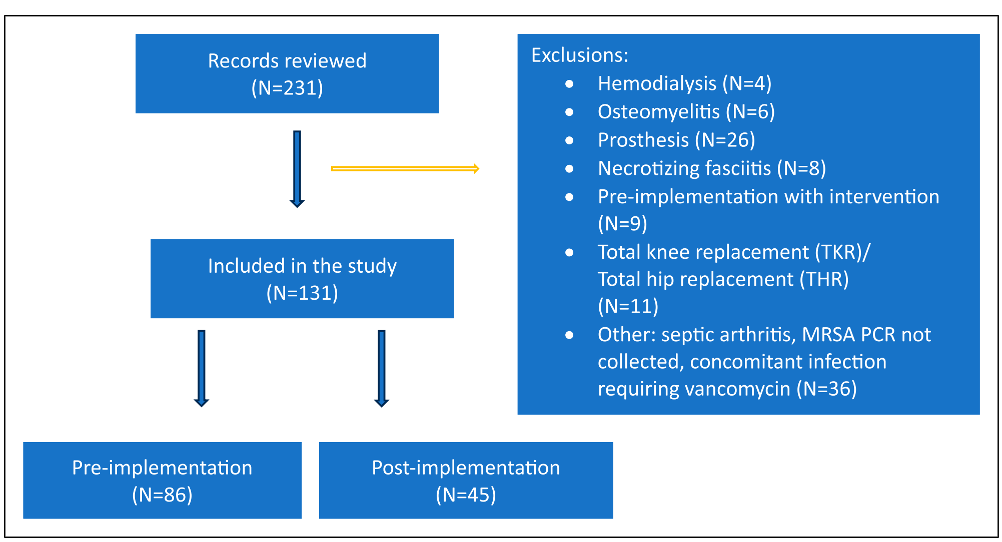

Figure 1. Inclusion and exclusion of study's participants.

Table 1. Demographics and Baseline Characteristics.

Baseline Variables	Pre- implementation (n = 86) No. (%) of patients	Post- implementation (n = 45) No. (%) of patients	*P* value
Age (years)			0.121
18- 30	7 (8.1)	2 (4.4)	
31-40	9 (10.5)	3 (6.7)	
41-50	14 (16.3)	12 (26.7)	
51-60	20 (23.3)	9 (20.0)	
61-70	16 (18.6)	3 (6.7)	
71-80	14 (16.3)	7 (15.6)	
Over 80	6 (7.0)	9 (20.0)	
Male	45 (52.3)	18 (40.0)	0.180
Antibiotic allergies	29 (33.7)	19 (42.2)	0.644
• Penicillin	10 (11.6)	8 (17.8)	
• Cephalosporins	6 (7.0)	1 (2.2)	
• Other antibiotics	13 (15.1)	10 (22.2)	
o TMP-SMX	9 (10.5)	7 (15.5)	
History of MRSA within last 12 months	8 (9.3)	3 (6.7)	0.605
Source			
• Skin	6 (6.9)	3 (100.0)	
• Urine	1 (1.2)	0	
• Unknown	1 (1.2)	0	
Hospitalization within 30 days of admission	15 (17.4)	1 (2.2)	0.011
Antibiotic in last 90 days	25 (29.1)	26 (57.8)	<0.001
• Oral	17 (19.8)	24 (53.3)	
• Intravenous	8 (9.3)	2 (4.4)	
No antibiotics in last 90 days	61 (70.9)	19 (42.2)	
IV Anti-MRSA agent	6 (7.0)	1 (2.2)	0.251
• Vancomycin	4 (4.7)	0	
• Clindamycin	1 (1.2)	1 (2.2)	
• SMX/TMP	1 (1.2)	0	
PO Anti-MRSA agent	12 (14.0)	14 (31.1)	0.019
• Clindamycin	2 (2.3)	3 (6.7)	
• Doxycycline	5 (5.8)	6 (13.3)	
• SMX/TMP	5 (5.8)	5 (11.1)	

34 (75.6%)]. The other most common types of infection were purulent cellulitis, abscesses, and wounds. Non-surgical cultures were more common than surgical cultures, 62.0% vs 76.5%, in the pre and post-group (Table 2).

Primary Outcomes and Secondary Outcomes

There was no difference between groups for the primary outcome; mean and standard deviation (SD) were 2.7 (1.9) and 2.6 (1.3), respectively, with a median LOT in days of 2.0 and 2.3, respectively, with a p-value 0.493 (Table 3). However, in the post-implementation group, 37 (82.2%) had a negative MRSA nasal PCR, and 8 (17.8%) had a positive MRSA nasal PCR. Mean and SD of vancomycin LOT in the negative and positive MRSA swab groups were 2.3 (1.1) and 3.9 (1.6) and this was found to be statistically significant with a p-value 0.006.

In the total cohort, 10 patients developed AKI with a mean vancomycin LOT of 3.3 days as compared to patients without AKI with a mean LOT of 2.6 days, p-value 0.644. Numerically, there were less AKIs in the-post implementation group, 2 vs 8 patients, but this was not found to be statistically significant, p-value 0.316. Nine patients were excluded from the serum creatinine assessment, with 4 of those receiving hemodialysis. There was no difference in the mean number of vancomycin levels in the pre- and post-implementation groups (1.4 vs 1.5, respectively, p-value 0.301). There was no difference in hospital LOS in the pre- and post-implementation groups, 5.5 and 5.4, respectively, p-value 0.269.

Discussion

To date, there are limited studies evaluating the clinical utility of a negative MRSA nasal PCR for de-escalation of

Table 2. Clinical Characteristics.

Clinical Variables	Pre- implementation (n = 86) No. (%) of patients	Post- implementation (n = 45)	*P*-value
Max temperature (°C) on admission day	37.28	37.18	0.305
Temperature 24hr (°C) after last dose or before discharge	36.87	36.75	0.057
Classification of SSTI			0.132
• Cellulitis non-purulent	44 (51.2)	34 (75.6)	
• Cellulitis, purulent	11 (12.8)	4 (8.9)	
• Abscess	13 (15.1)	2 (4.4)	
• Wound	11 (12.8)	3 (6.7)	
Location of infection			0.007
• Lower extremities	29 (33.7)	25 (55.6)	
• Upper extremities	24 (27.9)	8 (17.8)	
• Face or neck	1 (1.2)	5 (11.1)	
• Head	1 (1.2)	0	
• Pelvis/bottom	10 (11.6)	3 (6.7)	
• Trunk	21(24.4)	4 (8.9)	
ID consulted	34 (39.5)	8 (17.8)	0.011
Cultures collected	83 (96.5)	42 (93.3)	0.224
• Skin specimen	50 (58.1)	17 (37.8)	0.027
• Type of culture			0.278
o surgical	19 (38.0)	4 (23.5)	
o non-surgical	31 (62.0)	13 (76.5)	
Positive MRSA culture source	10 (11.6)	7 (15.6)	0.477
• Skin	9 (10.5)	7 (15.6)	0.388
• Blood	1 (1.2)	0	0.603
Other skin growth			
• MRSE	1 (1.2)	0	
• Corynebacterium	6 (7.0)	4 (8.9)	

Table 3. Primary and Secondary Outcomes.

Outcomes	Pre-implementation (n = 86)	Post-implementation (n = 45)	P = value
Primary			
Vancomycin LOT in days; mean (SD)	2.7 (1.9)	2.6 (1.3)	0.493
Secondary outcomes			
Number of vancomycin levels	1.4	1.5	0.301
Mean hospital length of stay in (days)	5.5	5.4	0.269
Incidence of AKI	8	2	0.316

vancomycin in SSTIs. Burgoon et al. conducted a similar retrospective analysis to determine the clinical utility of the MRSA nasal PCR for its potential effects on antimicrobial stewardship efforts.[4] However, their primary outcome was to compare the duration of vancomycin in the positive MRSA nasal PCRs vs the negative PCRs. Whereas, in our present study, we compared vancomycin LOT before and after the MRSA nasal PCR protocol was implemented. In our present study, we describe a distinctive expansion of pharmacists' scope of practice to include ordering of the MRSA nasal PCR based on a protocol and interpretations of results. In the Burgoon et al. study, the primary outcome of duration of vancomycin therapy was significantly longer in patients with a positive MRSA nasal PCR. Patients with a positive MRSA nasal PCR had a median of 4 days of vancomycin therapy [(interquartile range (IQR) 3-6], while patients with a negative MRSA nasal PCR had a median of 3 days of vancomycin therapy (IQR 2-5; *p-value* 0.01).[4] We found similar results, in the post-implementation group. Vancomycin mean LOT between the positive and negative MRSA PCR groups suggested a shorter LOT in the negative MRSA PCR group, 3.9 days vs 2.3 days, *p-value* 0.006. However, we did not see a statistically significant difference in vancomycin LOT when comparing the pre-implementation vs the post-implementation period. A reason that we may have not seen a difference in our primary outcome was that our post-implementation period was from January-September 2022, however our education for this intervention rolled out August 29th, 2022, and the BPA fired for the pharmacists to order the MRSA nasal PCR for SSTI indication on September 1st of 2022. This only left approximately a month for the prescribers to become acclimated to the protocol. If we had allowed adequate time post-protocol implementation, we may have been able to detect a difference in vancomycin LOT between groups. Patients in the post-implementation group received more anti-MRSA agents 90 days prior to admission (Table 1). We hypothesized that this could have led to a longer vancomycin LOT during the admission in the post-implementation group. However, longer vancomycin LOT could also be due to other reasons such as the severity of the wound condition, other clinical parameters, and whether the clinicians felt comfortable and confident to de-escalate based on the MRSA PCR result.

Although our study did not find a significant difference in the incidence of AKI between groups, patients who developed AKI had an average vancomycin LOT of 3 days, underlining the importance of de-escalating vancomycin as early as possible to prevent adverse effects. Additionally, we believe we did not see a difference in the other secondary outcomes, (vancomycin levels collected and hospital LOS), since we did not detect a difference in our primary outcomes between implementation groups.

A potential limitation of our study was that the retrospective, quasi-experimental design of our study did not involve a random assignment. Our study only incorporated patients at a single center teaching hospital. Our power was calculated with a 2:1 ratio including 95 patients in the pre-implementation group and 45 in the post-implementation group, however we did not meet power, as 10 patients were excluded from the pre-implementation group for having a MRSA nasal PCR obtained.

Conclusions

Among patients with SSTIs, a novel antimicrobial stewardship initiative utilizing the MRSA nasal PCR for vancomycin de-escalation did not significantly change the vancomycin LOT. However, in the post-implementation group, there was a significant difference in vancomycin LOT between negative and positive MRSA PCRs. No difference between groups in-regards to hospital LOS or mean number of vancomycin levels obtained was found.

Declaration of Conflicting Interests

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